# LittleHeart

AI heart murmur screening for children · UnivaBio 2026: AI for Human Health

## The problem

About 1 in 100 babies is born with a heart defect, and rheumatic heart disease affects tens of millions of people, most of them young and living in low- and middle-income countries. Both are often first noticed as a heart murmur. But recognising a murmur takes years of training, and the echocardiogram that confirms it is scarce exactly where these diseases are most common.

## Our solution

**LittleHeart** lets a community health worker, school nurse or GP record a child's heart at the four valve areas and instantly learn: *refer for echocardiogram*, *no murmur detected*, or *recording unclear, record again*, so scarce echocardiograms go to the children who need them.

## Key features

• **Guided recording**: step-by-step through the four sites with a live heartbeat and sound-level check.
• **Explainable results**: chest diagram, spectrogram per valve and a murmur-likelihood timeline.
• **Train your ear**: health workers practise on 18 real recordings against the AI.
• **Screening log** for school or village screening days, with CSV export.
• **Works offline**, installs like an app, and runs privately in the browser: no audio is uploaded.
• **4 languages** (English, Turkish, Portuguese, Spanish); real held-out cases; printable referral note.
• **Transparent validation tab** with an honest "working / not yet / limits" table.

## Honest limits

A screening aid, not a diagnosis or medical device. It misses some (mostly faint) murmurs, was trained on one population and one stethoscope, cannot tell innocent from pathological murmurs, and has not been tested with real phone adapters or prospectively. All of this is stated in the app.

## How the AI works

Trained on the CirCor DigiScope dataset (PhysioNet): 3,163 recordings from 942 Brazilian children, labelled by cardiologists. Each recording becomes a log-mel spectrogram (25-1000 Hz), cut into 4-second windows that a small convolutional network (about 61,000 weights, PyTorch, run in the browser with ONNX Runtime Web) scores as murmur / no murmur / unclear. During training, audio is randomly damaged with noise and phone-like filtering. A **heartbeat safety check** refuses recordings with no 40-220 bpm rhythm, so the AI never guesses on noise or speech.

## Results

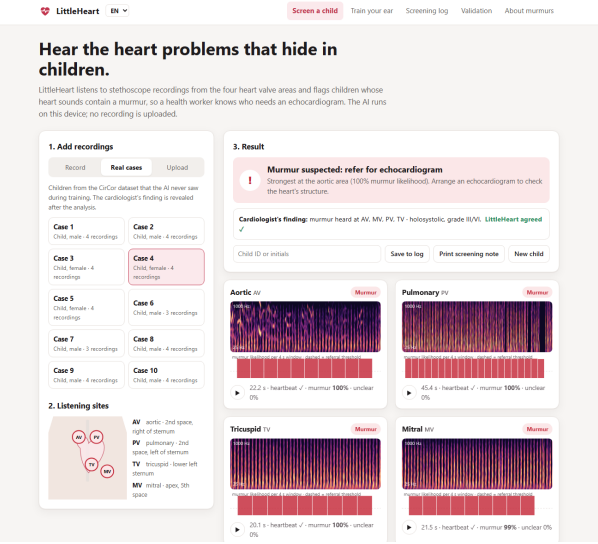
5-fold cross-validation, split by child, 932 children: PhysioNet 2022 murmur score **0.79 ± 0.02** (baseline 0.60; the best challenge team scored 0.78 on a harder hidden test set, not a like-for-like comparison). It caught **86%** of children with murmurs and cleared **91%** of those without (AUC 0.95). Of 24 murmurs missed, 22 were the faintest grade; loud grade III/VI murmurs were referred 100% of the time. With a phone-like microphone, training on randomly damaged audio raised murmurs caught from 36% to **83%**.

## External test

On the PhysioNet 2016 database (3,240 recordings, mostly adults, other devices, never used in training), the murmur score separated normal from abnormal hearts with AUC **0.57**, and only 75% of recordings passed the heartbeat check. Its labels are not murmurs, but the result shows LittleHeart must be validated locally before use in a new setting.

## Next steps

Test with low-cost stethoscope adapters, validate with children in a new region, add innocent-vs-pathological classification with echocardiography-labelled data, and pilot with a school RHD-screening programme.




Tech: Python, PyTorch, scikit-learn, NumPy (training) · HTML, CSS, JavaScript, ONNX Runtime Web (app) · Data: CirCor DigiScope (Oliveira et al. 2022) and PhysioNet/CinC Challenge 2016 (Liu et al. 2016), PhysioNet.